



Introduction to

Internet of Things

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: 15 X 1= 15

QUESTION 1:

What is the full form of IoT?

- a. Internet of Tasks
- b. Internet of Things**
- c. Internet of Tasks
- d. None of these

Correct Answer: b. Internet of Things

Detailed Solution: The full form of IoT is “Internet of Things”

See lecture 1 (Introduction to IoT – Part - I) @ 1:30

QUESTION 2:

Which of the following technologies have unified and has resulted in the evolution of IoT?

- a. Low-power embedded systems
- b. Cloud Computing
- c. Machine Learning
- d. All of these**

Correct Answer: d. All of these

Detailed Solution: Unification of technologies which has resulted in the advancement of IoT are –

- a. Low-power embedded systems
- b. Cloud Computing**
- c. Big Data
- d. Machine Learning**
- e. Networking**

See lecture 1 (Introduction to IoT – Part - I) @ 5:54



QUESTION 3:

Which of the following are the enablers of IoT?

- a. RFID
- b. Nanotechnology
- c. Sensors
- d. All of these

Correct Answer: d. All of these

Detailed Solution: The enablers of IoT are –

- a. RFID
- b. Nanotechnology
- c. Sensors
- d. Smart Networks

See lecture 2 (Introduction to IoT – Part - I) @ 12:50

QUESTION 4:

Which of the following is/are a function of an IoT Gateway?

- a. Connects the IoT LAN to a WAN
- b. Can implement several LAN and WAN
- c. Forwards packets between LAN and WAN on the IP layer
- d. All of these

Correct Answer: d. All of these

Detailed Solution: The functionalities of an IoT Gateway are –

- a. Connects the IoT LAN to a WAN
- b. Can implement several LAN and WAN
- c. Forwards packets between LAN and WAN on the IP layer

See lecture 2 (Introduction to IoT – Part - II) @ 3:09



QUESTION 5:

State whether the following statement is True or False.

Statement: “The same address of an IoT device may be repeated in the domain of another gateway. The gateway has a unique network prefix, which can be used to identify them globally.”

a. True

b. False

Correct Answer: a. True

Detailed Solution: The same address of an IoT device may be repeated in the domain of another gateway. The gateway has a unique network prefix, which can be used to identify them globally.

See lecture 2 (Introduction to IoT – Part - II) @ 07:00

QUESTION 6:

In _____ a node/network is connected to multiple networks for improved reliability.

a. Transparent roaming

b. Multi-homing

c. None of these

d. Both (a) and (b)

Correct Answer: b. Multi-homing

Detailed Solution: In multi-homing, a node/network is connected to multiple networks for improved reliability.

See lecture 2 (Introduction to IoT – Part - II) @ 15:22

QUESTION 7:

The IPv4 notation uses _____ number of bits to represent an address.

a. 33

b. 32

c. Both (a) and (b)

d. None of these

Correct Answer: b. 32



Detailed Solution: The IPv4 notation uses 32 bits to represent an address.

See lecture 2 (Introduction to IoT – Part - II) @ 16:33

QUESTION 8:

The “Source Address” field in the IPv6 header is of _____ bits in length.

- a. 32
- b. 64
- c. 128
- d. None of these

Correct Answer: c. 128

Detailed Solution: The IPv6 header uses 128 bits to represent the “Source Address”.

See lecture 2 (Introduction to IoT – Part - II) @ 18:03

QUESTION 9:

A device which detects or measures a physical property and records, indicates, or otherwise responds to it is called _____?

- a. A sensor
- b. An actuator
- c. A Transducer
- d. A processor

Correct Answer: a. A sensor

Detailed Solution: A device which detects or measures a physical property and records, indicates, or otherwise responds to it is called a sensor.

See lecture 3 (Sensing) @ 03:28

QUESTION 10:



A microphone is an example of an _____.

- a. Input device and actuator
- b. Only actuator
- c. Only Transducer
- d. Input device and transducer

Correct Answer: d. Input device and transducer

Detailed Solution: A microphone is an example of an input device and transducer.

See lecture 3 (Sensing) @ 10:31

QUESTION 11:

State whether the following statement is True or False.

Statement: A sensor is only sensitive to the measured property for which it has been made (e.g., A temperature sensor senses the ambient temperature of a room.)

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: A sensor is only sensitive to the measured property (e.g., A temperature sensor senses the ambient temperature of a room.)

See lecture 3 (Sensing) @ 12:11

QUESTION 12:

Based on the data type, sensors are classified as _____.

- a. Only scalar
- b. Only vector
- c. Scalar and Vector/Multimedia
- d. None of these

Correct Answer: c. Scalar and Vector/Multimedia

Detailed Solution: Based on the data type, sensors are classified as Scalar and Vector/Multimedia.

See lecture 3 (Sensing) @ 13:05



QUESTION 13:

State whether the following statement is True or False.

Statement: The more the resolution of a sensor, the less accurate its precision.

- a. True
- b. False

Correct Answer: b. False

Detailed Solution: The more the resolution of a sensor, the more accurate its precision.

See lecture 3 (Sensing) @ 12:46

QUESTION 14:

The sensitivity of a sensor under real conditions may differ from the value specified. This is called

- a. Physical error
- b. Sensitivity error
- c. Rounding error
- d. None of these

Correct Answer: b. Sensitivity error

Detailed Solution: The sensitivity of a sensor under real conditions may differ from the value specified.

See lecture 3 (Sensitivity error) @ 20:56

QUESTION 15:

A sensor node is made up of which of the following?

- a. Sensor/Sensing units
- b. A processing unit
- c. A power unit
- d. All of these

Correct Answer: d. All of these



Detailed Solution: A sensor node is made up of a combination of sensor/sensors, a processor unit, a radio unit, and a power unit.

See Page number – 101, Chapter - 5, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

*****END*****



**Introduction to
Internet of Things
Assignment-Week 2**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: 15 X 1= 15

QUESTION 1:

The full form of MQTT is

- a. Message Query Telemetry Transport
- b. Message Queue Telemetry Transport**
- c. Message Queue Telemedicine Transport
- d. None of these

Correct Answer: b. Message Queue Telemetry Transport

Detailed Solution: The full form of MQTT is Message Queue Telemetry Transport.

See lecture 6 (Basics of IoT Networking – Part II) @ 01:51

QUESTION 2:

In MQTT, a _____ controls the publish-subscribe messaging pattern.

- a. Message Broker**
- b. Publishers
- c. Subscribers
- d. All of these

Correct Answer: a. Message Broker

Detailed Solution: In MQTT, a message broker controls the publish-subscribe messaging pattern.



See lecture 6 (Basics of IoT Networking – Part II) @ 03:33

QUESTION 3:

Which of the following is NOT a Data Protocol in IoT?

- a. MQTT
- b. CoAP
- c. Websocket
- d. None of these

Correct Answer: d. None of these

Detailed Solution: The data protocols in IoT are –

- a) MQTT
- b) CoAP
- c) AMQP
- d) Websocket

See lecture 6 (Basics of IoT Networking – Part II) @ 01:16

QUESTION 4:

In MQTT for IoT, the Publishers are _____.

- a. Lightweight sensors
- b. Actuators
- c. Processing units
- d. None of these

Correct Answer: a. Lightweight sensors

Detailed Solution: In MQTT, the Publishers are lightweight sensors.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:49



QUESTION 5:

Which of the following is NOT a method in MQTT?

- a. Break
- b. Connect
- c. Disconnect
- d. None of these

Correct Answer: a. Break

Detailed Solution: Following are the methods in MQTT –

- a. Connect
- b. Disconnect
- c. Subscribe
- d. Unsubscribe
- e. Publish

See lecture 6 (Basics of IoT Networking – Part II) @ 06:18

QUESTION 6:

Which of the following protocols in IoT have been designed for Machine to Machine (M2M) applications such as smart energy and building automation?

- a. MQTT
- b. CoAP
- c. Websocket
- d. All of these

Correct Answer: b. CoAP

Detailed Solution: CoAP has been designed for Machine to Machine (M2M) applications such as smart energy and building automation.



See lecture 7 (Basics of IoT Networking – Part III) @ 00:41

QUESTION 7:

Similar to HTTP, CoAP utilizes which of the following for operation?

- a. GET
- b. PUT
- c. PUSH
- d. All of these

Correct Answer: d. All of these

Detailed Solution: Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE message requests to retrieve, create, update, and delete, respectively.

See lecture 7 (Basics of IoT Networking – Part III) @ 08:24

QUESTION 8:

What is the full form of XMPP in IoT?

- a. Extensible Markup Page Processing
- b. Extensible Messaging and Presence Protocol
- c. Both (a) and (b)
- d. Neither (a) nor (b)

Correct Answer: b. Extensible Messaging and Presence Protocol

Detailed Solution: XMPP stands for Extensible Messaging and Presence Protocol.

See lecture 7 (Basics of IoT Networking – Part III) @ 11:58

QUESTION 9:

State whether the following statement if True or False.



Statement: XMPP is an open standard protocol.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: XMPP is an open standard protocol.

See lecture 7 (Basics of IoT Networking – Part III) @ 11:58

QUESTION 10:

What is the full form AMQP in context of IoT protocol?

- a. Another Message Queuing Protocol
- b. Anchored Message Queuing Protocol
- c. Advanced Message Queuing Protocol
- d. None of these

Correct Answer: c. Advanced Message Queuing Protocol

Detailed Solution: AMQP stands for Advanced Message Queuing Protocol.

See lecture 8 (Basics of IoT Networking – Part IV) @ 0:41

QUESTION 11:

What is the basic unit of data in the AMQP protocol?

- a. Chunk
- b. Byte
- c. Frame
- d. None of these

Correct Answer: c. Frame

Detailed Solution: In AMQP, the basic unit of data is called a frame.



See lecture 8 (Basics of IoT Networking – Part IV) @ 0:41

QUESTION 12:

Which of the following is/are exchange types in AMQP?

- a. Direct
- b. Fan-out
- c. Topic
- d. All of these

Correct Answer: d. All of these

Detailed Solution: The AMQP exchange types are –

- a) Direct
- b) Fan-out
- c) Topic
- d) Header

See lecture 8 (Basics of IoT Networking – Part IV) @ 08:56

QUESTION 13:

IoT networks can be classified in which of the following type/types?



- a. Non-Beacon Enabled
- b. Beacon Enabled
- c. None of these
- d. Both (a) and (b)

Correct Answer: d. Both (a) and (b)

Detailed Solution: The IoT networks can be classified into two types such as

- i. Non-Beacon Enabled
- ii. Beacon Enabled

See lecture 9 (Connectivity Technologies – Part-I) @ 10:19

QUESTION 14:

The OSI model has _____ layers.

- a. 6
- b. 7
- c. 4
- d. 8

Correct Answer: b. 7

Detailed Solution: The OSI model is a conceptual framework that divides any networked communication system into seven layers.

See Page number – 10, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)



QUESTION 15:

The “Destination Address” in the IPv4 packet represents which of the following?

- a. The destination node address of the packet
- b. The intermediate hop in the network
- c. Both (a) and (b)
- d. None of these

Correct Answer: a. The destination node address of the packet

Detailed Solution: The “Destination Address” in the IPv4 packet represents the address of the destination node in the network.

See Page number – 18, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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**Introduction to
Internet of Things
Assignment-Week 3**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: 15 X 1= 15

QUESTION 1:

State whether the following statement is true or false.

Statement: There is no network layer in Wired HART

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: There is no network layer in Wired HART.

See lecture 11 (Connectivity Technologies-III) @ 05:22

QUESTION 2:

HART operates only in _____ GHz ISM band.

- a. 2.4
- b. 4.8
- c. 4.9
- d. 4.4

Correct Answer: a. 2.4

Detailed Solution: HART operates only in 2.4 GHz ISM band.



See lecture 11 (Connectivity Technologies-III) @ 05:43

QUESTION 3:

Super-frames in HART consist of grouped _____ wide timeslots.

- a. 10ms
- b. 20ms
- c. 55ms
- d. None of these

Correct Answer: a. 10ms

Detailed Solution: Super-frames in HART consist of grouped 10ms wide timeslots.

See lecture 11 (Connectivity Technologies-III) @ 07:05

QUESTION 4:

The process of identifying channels consistently affected by interference and removing them from use is known as _____.

- a. Channel hopping
- b. Channel aggregating
- c. Frequency aggregating
- d. Channel blacklisting

Correct Answer: d. Channel blacklisting

Detailed Solution: Channel Blacklisting identifies channels consistently affected by interference and removes them from use.

See lecture 11 (Connectivity Technologies-III) @ 07:30



QUESTION 5:

State whether the following statement is True or False

Statement: Each node is supervised by the application manager and it guides them on when and where to send packets.

- a. True
- b. False

Correct Answer: b. False

Detailed Solution: The Network manager supervises each node in the network and guides them on when and where to send packets.

See lecture 11 (Connectivity Technologies-III) @ 12:14

QUESTION 6:

FeliCa is commonly found in which country?

- a. Japan
- b. India
- c. USA
- d. None of these

Correct Answer: a. Japan

Detailed Solution: FeliCa is commonly found in Japan.

See lecture 11 (Connectivity Technologies-III) @ 17:52

QUESTION 7:

State whether the following state is true or false.

Statement: NFC tags found in supermarket products are examples of passive NFC.

- a. True



b. False

Correct Answer: a. True

Detailed Solution: Passive NFC tags found in supermarket products are examples of passive NFC.

See lecture 11 (Connectivity Technologies-III) @ 18:47

QUESTION 8:

Active NFC devices are able to _____?

- a. transmit information
- b. collect information
- c. Both (a) and (b)
- d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: Active NFC devices are able to collect as well as transmit information.

See lecture 11 (Connectivity Technologies-III) @ 18:52

QUESTION 9:

A _____ in NFC emits a small electric current which creates a magnetic field that in turn bridges the physical space between the devices.

- a. reader
- b. writer
- c. destroyer
- d. None of these

Correct Answer: a. reader



Detailed Solution: A reader in NFC emits a small electric current which creates a magnetic field that in turn bridges the physical space between the devices

See lecture 11 (Connectivity Technologies-III) @ 20:31

QUESTION 10:

Bluetooth technology is based on _____.

- a. WiFi
- b. LoRa
- c. Ad-hoc technology
- d. None of these

Correct Answer: c. Ad-hoc technology

Detailed Solution: Bluetooth technology is based on Ad-hoc technology.

See lecture 12 (Connectivity Technologies-IV) @ 02:30

QUESTION 11:

Bluetooth technology operates in the unlicensed industrial, scientific and medical (ISM) band at _____ to _____ GHz.

- a. 3.2, 5.6
- b. 2.4, 2.485
- c. 5.0, 5.89
- d. None of these

Correct Answer: b. 2.4,2.485

Detailed Solution: Bluetooth technology operates in the unlicensed industrial, scientific and medical (ISM) band at 2.4 to 2.485 GHz.

See lecture 12 (Connectivity Technologies-IV) @ 04:39



QUESTION 12:

Z-wave operates at _____ MHz in the US and _____ MHz in Europe.

- a. 1008, 989
- b. 908.42, 868.42
- c. 767.56, 855.28
- d. None of these

Correct Answer: b. 908.42, 868.42

Detailed Solution: Z-wave operates at 908.42 MHz in the US and 868.42 MHz in Europe.

See lecture 13 (Connectivity Technologies-V) @ 3:40

QUESTION 13:

State whether the following statement is true or false.

Statement: Sensor nodes have limited battery life.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: Sensors nodes have limited battery life.

See lecture 14 (Sensor Networks-I) @ 13:15

QUESTION 14:

State whether the following statement is true or false.



Statement: The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

See Page number – 157, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 15:

Zigbee commonly uses _____ data rate.

- a. 250 kbps
- b. 250 Mbps
- c. 260 kbps
- d. 260 Mbps

Correct Answer: a. 250 kbps

Detailed Solution: Zigbee commonly uses 250 kbps data rate.

See Page number – 131, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)



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**Introduction to
Internet of Things
Assignment-Week 4**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15**

Total marks: 15 X 1= 15

QUESTION 1:

State whether the following statement is true or false.

Statement: Coverage in WSN is defined as the area-of-interest that is covered satisfactorily.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: Coverage in WSN is defined as the area-of-interest that is covered satisfactorily.

See lecture 17 (Sensor Networks-IV) @ 01:44

QUESTION 2:

If transmission range $\geq 2 \times$ sensing range,

- a. Coverage implies connectivity
- b. Coverage implies greater coverage
- c. Both (a) and (b)
- d. None of these

Correct Answer: a. Coverage implies connectivity



Detailed Solution: If transmission range $\geq 2 \times$ sensing range, coverage implies connectivity.

See lecture 17 (Sensor Networks-IV) @ 04:14

QUESTION 3:

Forest fire monitoring in WSN is an example of which type of reporting?

- a. Event driven
- b. On-demand
- c. Scattered
- d. None of these

Correct Answer: a. Event driven

Detailed Solution: Forest fire monitoring in WSN is an example of event driven reporting.

See lecture 17 (Sensor Networks-IV) @ 08:13

QUESTION 4:

State whether the following statement is true or false.

Statement: Objective of coverage in WSN is to use minimum number of sensors and maximize network lifetime.

- a. True
- b. False

Correct Answer: a. True



Detailed Solution: The objective of coverage in WSN is to use minimum number of sensors and maximize the network lifetime.

See lecture 17 (Sensor Networks-IV) @ 08:47

QUESTION 5:

What is the full form of AUV in the context of mobile wireless sensor networks?

- a. Aerial and Underwater Vehicle
- b. Ambient and Underprivileged Vehicle
- c. Autonomous Underwater Vehicle
- d. None of these

Correct Answer: c. Autonomous Underwater Vehicle

Detailed Solution: The full form of AUV is Autonomous Underwater Vehicle.

See lecture 18 (Sensor Networks-V) @ 07:56

QUESTION 6:

In mobile wireless sensor networks, the sensor nodes are capable of _____ and measuring the condition of their surrounding environments.

- a. collaborating with one another
- b. destroying one another
- c. Both (a) and (b)
- d. None of these

Correct Answer: a. collaborating with one another

Detailed Solution: In mobile wireless sensor networks, the sensor nodes are capable of collaborating with one another and measuring the condition of their surrounding environments.



See lecture 18 (Sensor Networks-V) @ 04:56

QUESTION 7:

State whether the following statement is true or false.

Statement: In mobile sensor networks, the sensor nodes come in close proximity of sink and deliver the data.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: In mobile sensor networks, the sensor nodes come in close proximity of sink and deliver the data.

See lecture 18 (Sensor Networks-V) @ 06:19

QUESTION 8:

Which of the following is/are a challenge/challenges in Human-centric Sensing?

- a. Energy of devices
- b. Participant selection
- c. Both (a) and (b)
- d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: The following are the challenges in human-centric sensing -

- a. Energy of devices
- b. Participant selection
- c. Privacy of users

See lecture 18 (Sensor Networks-V) @ 12:45

QUESTION 9:



Which of the following is/are a feature/features of UAV networks?

- a. Mesh or star networks
- b. Multi-tasking
- c. Large coverage area
- d. All of these

Correct Answer: d. All of these

Detailed Solution: The following are the features of UAV networks -

- d. Mesh or star networks
- e. Multi-tasking
- f. Large coverage area

See lecture 19 (UAV Networks) @ 02:43

QUESTION 10:

State whether the following statement is true or false.

Statement: Frequently change in network topology is a benefit of UAV networks.

- a. False
- b. True

Correct Answer: a. False

Detailed Solution: Frequently change in network topology an issue of UAV networks.

See lecture 19 (UAV Networks) @ 05:29

QUESTION 11:

Which of the following is NOT a feature of multi-UAV system?

- a. Low failures



- b. High scalability
- c. Low survivability
- d. High cost

Correct Answer: c. Low survivability

Detailed Solution: Following are some of the features of multi-UAV system –

- a. Low failures
- b. High scalability
- c. High survivability
- d. High cost

See lecture 19 (UAV Networks) @ 07:53

QUESTION 12:

State whether the following statement is true or false.

Statement: M2M communications are free of any human intervention.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: M2M communications are free of any human intervention.

See lecture 20 (Machine to Machine Communication) @ 05:44



QUESTION 13:

Which of the following is NOT an M2M node type?

- a. Secure End Node
- b. High End Node
- c. Low End Mode
- d. Mid End Node

Correct Answer: a. Secure End Node

Detailed Solution: M2M node types are as follows –

- a. Low End Node
- b. Mid End Node
- c. High End Node

See lecture 20 (Machine to Machine Communication) @ 10:43

QUESTION 14:

What is the full form of M2SP in the context of Machine-to-Machine Communication?

- a. M2M Service Platform
- b. M2M Session Packet
- c. M2M Secure Packaging
- d. None of these

Correct Answer: a. M2M Service Platform

Detailed Solution: M2SP stands for M2M Service Platform.

See lecture 20 (Machine to Machine Communication) @ 16:31

QUESTION 15:

The _____ provides integrated services based on device collected data-sets.



- a. M2M Application Platform
- b. M2M Service Platform
- c. M2M Hardware Platform
- d. None of these

Correct Answer: a. M2M Application Platform

Detailed Solution: The M2M Application Platform provides integrated services based on device collected data-sets.

See lecture 20 (Machine to Machine Communication) @ 18:00

*****END*****



**Introduction to
Internet of Things
Assignment-Week 5**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: 15 X 1= 15

QUESTION 1:

Which of the following is/are current challenge/challenges in IoT?

- a. Large scale of co-operation
- b. Global heterogeneity
- c. Unknown IoT device configuration
- d. All of these**

Correct Answer: d. All of these

Detailed Solution: Current challenges in IoT are –

- a. Large scale of co-operation
- b. Global heterogeneity
- c. Unknown IoT device configuration
- d. Semantic conflicts

See lecture 21 (Interoperability in Internet of Things) @ 03:41.



QUESTION 2:

When two IoT devices wish to communicate, semantic conflict in such a scenario in IoT interoperability refers to_____.

- a. Two devices built by the same manufacturer
- b. Two devices having different processing and business logic
- c. Two devices sensing the same physical parameter
- d. Two devices having different deployment location

Correct Answer: b. Two devices having different processing and business logic

Detailed Solution: Semantic conflict refers to when different IoT devices have different processing and business execution logic. Refer Lecture 21@5:32

QUESTION 3:

Which of the following issues needs to be addressed while solving user interoperability?

- a. Device characterization and identification
- b. Syntactic interoperability
- c. Both (a) and (b)
- d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: All of the options given should be addressed while solving user interoperability issues. Refer Lecture 21@15:16.

QUESTION 4:



State whether the following statement is true or false

Statement: UMB core interoperability component is responsible for converting physical devices into virtually abstracted ones.

a. False

b. True

Correct Answer: a. False

Detailed Solution: UMB-A is responsible for converting physical devices into virtually abstracted ones.
(Please refer Lecture 21@27:16)

QUESTION 5:

What is the full form of UMB in IoT interoperability?

a. Universal Middleware Bridge

b. Universal Main Bridge

c. Universal Main Bracket

d. None of these

Correct Answer: a. Universal Middleware Bridge

Detailed Solution: UMB stands for Universal Middleware Bridge.

Refer Lecture 21@22:16.

QUESTION 6:



Which of the following is NOT a function available in the Servo library for Arduino ?

- a. kill()
- b. destroy()
- c. burn()
- d. All of these

Correct Answer: d. All of these

Detailed Solution: The following are NOT functions of the Servo library for Arduino –

- a. kill()
- b. destroy()
- c. burn()

The Refer Lecture 25@18:08.

QUESTION 7:

Arduino UNO Board is open source?

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: An Arduino UNO board is an open source electronic programmable board.

Refer Lecture 22@05:21.



QUESTION 8:

Arduino Uno board accepts which type of input/inputs?

- a. Analog
- b. Digital
- c. Both (a) and (b)
- d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: Arduino UNO board accepts analog and digital signals as input.

Refer Lecture 22@05:21

QUESTION 9:

Arduino boards are based on which of the following microcontroller/microcontrollers.

- a. ATMEGA328
- b. ATMEGA32u4
- c. Both (a) and (b)
- d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: Arduino board are based on the following microcontrollers –

- a. ATMEGA328
- b. ATMEGA32u4
- c. ATMEGA2560
- d. AT91SAM3X8E



Refer Lecture 22@06:20

QUESTION 10:

What is the clock speed of the Arduino Uno board?

- a. 16MHz
- b. 17MHz
- c. 8MHz
- d. None of these

Correct Answer: a. 16MHz

Detailed Solution: The clock speed of Arduino Uno board is 16MHz.

Refer Lecture 22 @ 07:13.

QUESTION 11:

Consider the following Arduino sketch.

```
int r = 4;
int g = 5;
void setup(){

Serial.begin(9600);
pinMode(r,INPUT);
pinMode(g,??);
}

void loop()
{
  int val = analogRead(r);
  if( val > 10)
  {
    digitalWrite(g,HIGH);
  }
  else
  {
    digitalWrite(g,HIGH);
  }
  delay(500);
}
```



What must be inserted in the place of ‘??’ within the second pinMode() function in void setup()?

- a. INPUT
- b. OUTPUT**
- c. None of these
- d. Anyone of these is okay

Correct Answer: b. OUTPUT

Detailed Solution: Since pin ‘g’ is writing digital values, it is required to set in OUTPUT mode. Refer Lecture 22-23.

QUESTION 12:

The “Verify” option in the Arduino IDE checks the code for _____.

- a. Compilation errors**
- b. Improper hardware connection
- c. Both (a) and (b)
- d. None of these

Correct Answer: a. Compilation errors

Detailed Solution: The “Verify” option in the Arduino IDE checks the code for compilation errors. Refer Lecture 24.

QUESTION 13:

How many pins does the DHT Digital Humidity and Temperature sensor have?



- a. 2
- b. 3
- c. 4
- d. None of these

Correct Answer: c. 4

Detailed Solution: DHT Digital Humidity and Temperature sensor has 4 pins.

See lecture 24 @ 03:23

QUESTION 14:

What is the function of the delay() function in an Arduino program?

- a. Initializes a sensor
- b. Introduces a delay for the specified time
- c. Stores the sensor value
- d. None of the above

Correct Answer: b. Introduces a delay for the specified time

Detailed Solution: The delay() function introduces a delay in the operation of the microcontroller for the specified duration.

See lecture 24 @13:51

QUESTION 15:

What does the following code snippet do?

Servo myservo;

- a. Creates a variable to store the sensor value
- b. Instructs the sensor to sense data
- c. Creates an instance of servo to use it in the Arduino sketch
- d. None of these



Correct Answer: c. Creates an instance of servo to use it in the Arduino sketch

Detailed Solution: The code snippet “Servo myservo;” creates an instance of servo to use in the Arduino sketch.

Refer Lecture 25 @5:32

*****END*****



**Introduction to
Internet of Things
Assignment-Week 6**

TYPE OF QUESTION:MCQ/MSQ

Number of questions: **15**

Total marks: 15 X 1= 15

QUESTION 1:

Python can only be used to build small applications.

- a. True
- b. False**

Correct Answer: b. False

Detailed Solution: With open-source nature, python forms a strong backbone to build large applications.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 3:46)

QUESTION 2:

Which of the following is NOT a Python IDE?

- a. Spyder
- b. PyCharm
- c. Both (a) and (b)
- d. None of these**



Correct Answer: d. None of these

Detailed Solution: Spyder and PyCharm are examples of Python IDE.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 05:12)

QUESTION 3:

State whether the following statement is true or false.

Statement: To indicate different blocks of code, python follows rigid indention.

- a. False
- b. True**

Correct Answer: b. True

Detailed Solution: To indicate different blocks of code, python follows rigid indention
(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 07:25).

QUESTION 4:

Which of the following is not a data-type in Python?

- a. List
- b. Branch**
- c. Tuple
- d. None of these

Correct Answer: b. Branch

Detailed Solution: The data-types in python are –



- a. List
- b. Tuple
- c. Dictionary

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 15:20).

QUESTION 5:

State whether the following statement is true or false.

Statement: Python allows us to read and write files.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: Python allows us to read and write files.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 01:17)

QUESTION 6:

Which of the following is NOT a mode to open a file?

- a. Read mode
- b. Write mode
- c. Append mode
- d. None of these

Correct Answer: d. None of these



Detailed Solution: The four basic modes to open a file in python are –

- a. Read mode
- b. Write mode
- c. Append mode
- d. Both read and write mode

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 03:00)

QUESTION 7:

State whether the following command to install the PIL library in Linux is correct or not.

sudo pip install pillow

- a. Correct
- b. Incorrect

Correct Answer: a. Correct

Detailed Solution: The command to install the PIL library is *sudo pip install pillow*.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 17:40)

QUESTION 8:

State whether the following statement is correct for creating a socket 's' in python.

s = socket.socket(socket_family, socket_type, protocol=0)

- a. Correct
- b. Incorrect

Correct Answer: a. Correct

Detailed Solution: The statement *s = socket.socket(socket_family, socket_type,*



protocol=0) creates a socket in python.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @24:27).

QUESTION 9:

What is a Raspberry Pi?

- a. A computer on your palm
- b. Single board computer
- c. Low cost
- d. All of these

Correct Answer: d. All of these

Detailed Solution: Raspberry pi is –

- a. A computer on your palm
- b. Single board computer
- c. Low cost
- d. Easy to access

(Please refer to lecture INTRODUCTION TO RASPBERRY PI-I @05:00).

QUESTION 10:

Which of the following are required for a basic set up of Raspberry Pi?

- a. Monitor
- b. Keyboard
- c. Mouse
- d. All of these



Correct Answer: d. All of these

Detailed Solution: The following are required for a basic set up of Raspberry Pi –

- a. Monitor
- b. Keyboard
- c. Mouse

See lecture INTRODUCTION TO RASPBERRY PI-I @ 11:33

QUESTION 11:

Which of the following keyboard key combination writes (i.e saves) the code to the file in nano editor ?

- a. Ctrl+O
- b. Ctrl+X
- c. Both (a) and (b)
- d. None of these

Correct Answer: a. Ctrl+O

Detailed Solution: Ctrl+O writes the code to the file in nano editor.

See lecture INTRODUCTION TO RASPBERRY PI-II @ 10:44

QUESTION 12:

A relay module for Raspberry Pi has how many terminals ?

- a. 3
- b. 2
- c. 4
- d. 5

Correct Answer: a. 3



Detailed Solution: A relay module for Raspberry Pi has 3 terminals.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-I @ 06:01.

QUESTION 13:

State whether the following statement is true or false.

Statement: Adafruit provides a library to work with the DHT22 sensor.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Adafruit provides a library to work with the DHT22 sensor.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-I @ 09:19

QUESTION 14:

What is the function of the close() function in python programming while handling files.

a. This function does not exist in python

b. It closes the current connection to the file and ensures that the file is free to use for other resources

c. None of these

d. Both (a) and (b)



Correct Answer: b. It closes the current connection to the file and ensures that the file is free to use for other resources

Detailed Solution: The close() function in python ensures that the file is free to use for other resources.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 4:44)

QUESTION 15:

Where is the Raspbian image found?

- a. It is not available on the Internet
- b. It is available as image file from the Raspberry Pi official website**
- c. It can only be purchased from offline vendor
- d. None of these

Correct Answer: b. It is available as image file from the Raspberry Pi official website

Detailed Solution: Raspbian image can be downloaded from the Raspberry Pi official website.

(Please refer to lecture INTRODUCTION TO RASPBERRY PI-I @ 12:32)

*****END*****